

Introduction to Database Systems

Homework 1: SQL

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1 Instructions

Write SQL queries to retrieve the information requested below from your PostgreSQL database. Then enter each query, as well as the numerical answer to each question, in LearnIT. The query should not return anything except the answer. As this is a learning exercise, you should take time to test variants and make the queries as simple as you possibly can. The queries should also be well and consistently formatted, and as readable as possible, as SQL queries are generally part of your code base.

Although you can use subqueries, you should not answer the question using a sequence of several queries, and you should not use any “magic constants” in your queries. Note that in many cases, some (incorrect) variants of some queries could return the same results as a correct query. The fact that a query returns the correct answer thus does not necessarily mean that the query itself is correct. Therefore, you must take care to insert additional data to test your queries adequately. In short, the query must work for **any instance of the schema**.

All the queries ask you to count the rows that satisfy some criteria. You should start by first creating a query that returns all rows that satisfy the given criteria. Once your query is ready, you can then turn it into a counting query, either by replacing the output columns with a `count` statement, or by enveloping the query with a counting query:

```
select count(*)
from (
  select ...
) tmp;
```

In the exam, you will also get directions along the following lines (but given that this is not an exam, you need not worry about partial credit): Queries must work for any database instance and should avoid system-specific language features (including `LIMIT 1`). Queries should not return anything except the answer; a query that returns more information will not receive full points, even if the answer is part of the returned result. A sequence of several queries that answer the question will not receive full points, but subqueries and

views can be used. Queries should be as simple as possible; queries that are unnecessarily complex may not get full points, despite returning the correct answer. If you are unable to complete the query, you can still submit your attempt, along with a brief description of the problem with the query, and you might still get partial points.

2 Database description

In this homework, you will work with the database `imdb`. To start working with the database, you must first create it and then run the `HW1-DB.sql` script available on LearnIT using `psql` on your laptop, as discussed in class:

```
psql [-U postgres] [-q] your_database_name < HW1-DB.sql
```

The database contains information on movies and people. Most of the data is actually part of the real IMDb database, but some information is fabricated. The following is the relevant part of the database schema (the relation `movie` has some additional attributes that are not used in the homework).

```
person(id, name, gender, birthdate, deathdate, height)
movie(id, title, year, color, country, language, releaseDate, ...)
genre(genre, category)
movie_genre(movieId, genre)
role(role)
involved(personId, movieId, role)
```

All relations have a defined primary key (all underlined attributes) except `movie_genre`. In particular, the `involved` relation has a composite primary key consisting of all three attributes. Foreign keys are defined where appropriate. The `role` relation is a lookup table with a list of all possible roles. The meaning of other relations and attributes should be self-explanatory.

3 Passing grade

The maximum grade is 40 points. You get 4 points for each correct SQL query and 1 point for each correct numerical answer. To pass the homework, you need to have *at least 8/40* points. However, the more you do, the more feedback you get and the more you learn.

4 Queries

In the following, you are given the correct answer to a very similar query but not the correct answer to the query you should produce in the end. Start by finding the correct form for the given answer and then convert the query to get the requested answer. As outlined above, each query should only return a single numerical answer. Answer each of the following questions using a single SQL query on the `imdb` database:

- (1) The `person` relation contains 284 entries with a registered death date after ‘2010-02-01’. How many entries do not have a registered death date?
- (2) In the database, there are 46 movies in the French language for which the average height of all the people involved is greater than 185 centimeters (ignoring people with unregistered height). What is the number of movies in the Portuguese language for which the average height of all people involved is greater than 175 centimeters?
- (3) The `movie_genre` relation does not have a primary key, which can lead to a movie having more than one entry with the same genre. As a result, there are 14 movies in `movie_genre` that have the genre ‘Action’ assigned to them more than once. How many movies in `movie_genre` have the genre ‘Thriller’ assigned to them more than once?
- (4) According to the information in the database, 52 different people acted in movies directed by ‘Ingmar Bergman’. How many different people acted in movies directed by ‘Akira Kurosawa’?
- (5) Of all the movies produced in 2007, there are 15 that have two directors `involved` in them. How many movies produced in 2010 have two directors `involved` in them?
- (6) There are 406 unique pairs of actors who have appeared together in exactly 10 movies released between 2000 and 2010. How many unique pairs of actors have appeared together in exactly 20 movies released between 2000 and 2010?
- (7) Of all the movies produced between 2000 and 2002, there are 782 that have entries registered in `involved` for *all* roles defined in the `roles` relation. How many movies produced between 2002 and 2004 have entries registered in `involved` for all roles defined in the `roles` relation? *Note: This is a relational division query that must work for any instance; Do not use any ‘magic numbers’.*
- (8) The number of people who have played a role in movies of *all* genres in the category ‘Newsworthy’ is 156. How many people have played a role in movies of *all* genres in the category ‘Newsworthy’ but have not played any role in movies that cover all genres in the category ‘Popular’?

HW1 SOLUTIONS FALL 2024

1. 1) SQL

ESSAY 4 points 0.10 penalty monospaced

1) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

2. 1) Numerical answer

NUMERICAL 1 point 0.10 penalty

1) Run the query of the previous question and paste the result here (an integer):

- 52759 ✓

3. 2) SQL

ESSAY 4 points 0.10 penalty monospaced

2) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

4. 2) Numerical answer

NUMERICAL 1 point 0.10 penalty

2) Run the query of the previous question and paste the result here (an integer):

- 26 ✓

5. 3) SQL

ESSAY 4 points 0.10 penalty monospaced

3) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

6. 3) Numerical answer

NUMERICAL 1 point 0.10 penalty

3) Run the query of the previous question and paste the result here (an integer):

- 14 ✓

7. 4) SQL

ESSAY 4 points 0.10 penalty monospaced

4) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

8. 4) Numerical answer

NUMERICAL 1 point 0.10 penalty

4) Run the query of the previous question and paste the result here (an integer):

- 129 ✓

9. 5) SQL

ESSAY 4 points 0.10 penalty monospaced

5) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

10. 5) Numerical answer

NUMERICAL 1 point 0.10 penalty

5) Run the query of the previous question and paste the result here (an integer):

- 9 ✓

11. 6) SQL

ESSAY 4 points 0.10 penalty monospaced

6) Write your SQL query here:

Notes for grader:

- See HW1-SOL-Fall-2024 file.

12. 6) Numerical answer

NUMERICAL 1 point 0.10 penalty

6) Run the query of the previous question and paste the result here (an integer):

- 4 ✓

13. **7) SQL**

ESSAY 4 points 0.10 penalty monospaced

7) Write your SQL query here:

Notes for grader:

- See HW1-SQL-Fall-2024 file.

14. **7) Numerical answer**

NUMERICAL 1 point 0.10 penalty

7) Run the query of the previous question and paste the result here (an integer):

- 823 ✓

15. **8) SQL**

ESSAY 4 points 0.10 penalty monospaced

8) Write your SQL query here:

Notes for grader:

- See HW1-SQL-Fall-2024 file.

16. **8) Numerical answer**

NUMERICAL 1 point 0.10 penalty

8) Run the query of the previous question and paste the result here (an integer):

- 48 ✓

Total of marks: 40